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PROPERTIES OF TRIANGLE

NIMCET PYQ

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By Amit Katiyar (MCA-JNU)

1. The angles of a triangle are in A.P. if $b : c = \sqrt{3}/\sqrt{2}$ then angle A is equal to:

NIMCET 2007

(a)$75°$ (b)$105°$
(c)$30°$ (d)$45°$

2. If a, b, c are in A.P., then $\tan\frac{A}{2}\tan\frac{C}{2} =$

NIMCET 2007

(a)$\frac{1}{3},\frac{2}{3}$ (b)2/3
(c)3/2 (d)none

3. In a $\Delta ABC, R$ is circumradius and $R^2 = a^2 + b^2 + c^2$, then the ΔABC is

NIMCET 2010

(a)Acute angled (b)Obtuse angled
(c)Right angled (d)None of these

4. The greatest angle of the triangle whose sides are $x^2 + x + 1, 2x + 1, x^2 - 1$ is

NIMCET 2011

(a)$150°$ (b)$90°$ (c)$135°$ (d)$120°$

5. If ABC is a triangle whose area is $10\sqrt{3}$ unit with side lengths $|AB| = 8$ unts and $|AC| = 5$ units, then the possible value of $\angle A$ is

NIMCET 2013

(a)$60°$ or $120°$ (b)$45°$ or $135°$
(c)only $30°$ (d)only $90°$

6. In a ΔABC, if $(c + a + b)(a + b - c) = ab$, then the measure of $\angle C$ is

NIMCET 2013

(a)$\frac{\pi}{3}$ (b)$\frac{\pi}{6}$ (c)$\frac{2\pi}{3}$ (d)None

7. If A, B and C are three angles of a ΔABC, whose area is Δ. Let a, b and c be the sides opposite to the angles, A, B and C, respectively. If $s = \frac{a+b+c}{2} = 6$, then the product $\frac{1}{3}s^2(s - a)(s - b)(s - c)$ is equal to

NIMCET 2014

(a)2Δ (b)$2\Delta^2$ (c)$\sqrt{2}\Delta$ (d)Δ^2

8. In ΔABC, if $a = 2, b = 4$ and $\angle C = 60°$, then A and B are respectively equal to

NIMCET 2014

(a)$90°, 30°$ (b)$45°, 75°$
(c)$60°, 60°$ (d)$30°, 90°$

9. If the angles of a triangle are in the ratio $2 : 3 : 7$, then the ratio of the sides opposite to these angles is

NIMCET 2015

(a)$\sqrt{2} : 2 : \sqrt{3} + 1$ (b)$2 ; \sqrt{2} : \sqrt{3} + 1$
(c)$2 : \sqrt{2} : \frac{\sqrt{2}}{\sqrt{3}-1}$ (d)$\frac{1}{\sqrt{2}} : 2 : \frac{\sqrt{3}+1}{2}$

10. In a $\Delta ABC, a = 4, b = 3, \angle BAC = 60°$, then the equation for which c is the root, is

NIMCET 2016

(a)$c^2 + 3c + 7 = 0$ (b)$c^2 + 3c - 7 = 0$
(c)$c^2 - 3c + 7 = 0$ (d)$c^2 - 3c - 7 = 0$

11. In a ΔABC, let $\angle C = \frac{\pi}{2}$. If r is the in radius and R is circumradius of the ΔABC, then $2(r + R)$ equals to

NIMCET 2017

(a)$a + c$ (b)$a + b + c$ (c)$a + b$ (d)$b + c$

12. If $\Delta = a^2 - (b - c)^2$, where Δ is the area of the ΔABC, then $\tan A$ equals to

NIMCET 2019

(a)$\frac{15}{16}$ (b)$\frac{8}{15}$ (c)$\frac{8}{17}$ (d)$\frac{1}{2}$

13. In a triangle, if the sum of two sides is x and their product is y such that $(x + z)(x - z) = y$, where z is the third side of the triangle, then the triangle is

NIMCET 2021

(a)Equilateral (b)Right angled
(c)Isosceles (d)Obtuse angled

14. If in a ΔABC a $\cos^2\frac{C}{2} + c\cos^2\frac{A}{2} = \frac{3b}{2}$, then the sides of triangle are in

NIMCET 2021

(a)AP (b)HP (c)GP (d)None

15. In a ΔABC, if the tangent to half the difference of two angles is equal to one third of the tangent of the sum of half the angles, then the ratio of the sides opposite to the angle is

NIMCET 2022

(a)2 : 1 (b)1 : 2 (c)3 : 1 (d)1 : 1

16. The perimeter of a ΔABC is 6 times the arithmetic mean of the sines of its angles. If the side a is 1, then the angle A

NIMCET 2023

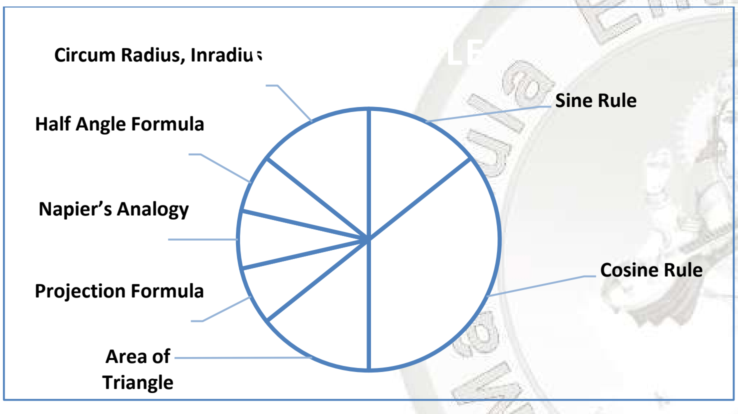
(a)$\frac{\pi}{6}$ (b)$\frac{\pi}{3}$ (c)$\frac{\pi}{2}$ (d)π

ANSWER

1.	a	2.	d	3.	c	4.	d	5.	a
6.	c	7.	b	8.	d	9.	a	10.	d
11.	c	12.	b	13.	d	14.	a	15.	a
16.	a								

Analysis 2008 to 2025
Questions Asked in NIMCET 2008 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	02	NIMCET 2016	01
NIMCET 2008	00	NIMCET 2017	01
NIMCET 2009	00	NIMCET 2018	00
NIMCET 2010	01	NIMCET 2019	01
NIMCET 2011	01	NIMCET 2020	00
NIMCET 2012	00	NIMCET 2021	02
NIMCET 2013	02	NIMCET 2022	01
NIMCET 2014	02	NIMCET 2023	01
NIMCET 2015	01	NIMCET 2024	00
		NIMCET 2025	00



Sub Topicwise Questions Asked in NIMCET

Sub Topic	Ques. Asked	Sub Topic	Ques. Asked
Arithmetic Progression	08	Sub Topic	
Geometric Progression	08	Sum of Miscellaneous Series	03
Harmonic Progression	07		
AP, GP, HP	06		
AM, GM, HM	06		
Sum of Squares of Natural Number	03		

SOLUTION

1. (a)

2. (d)

3. (c)

Given, $8R^2 = a^2 + b^2 + c^2$

In a $\triangle ABC, 2R = \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

$2R\sin A = a, 2R\sin B = b, 2R\sin C = c$

$\therefore\ 8R^2 = 4R^2[\sin^2 A + \sin^2 B + \sin^2 C]$

$\sin^2 A + \sin^2 B + \sin^2 C = 2$

$\sin^2 C = (1 - \sin^2 A) + (1 - \sin^2 B)$

$\sin^2 C = \cos^2 A + \cos^2 B$

$\because\ A + B + C = \pi \Rightarrow C = \pi - (A + B)$

$\sin C = \sin[\pi - (A + B)] = \sin(A + B)$

From Eq. (i)

$[\sin(A + B)]^2 = \cos^2 A + \cos^2 B$

$\Rightarrow\ (\sin A\cos B + \cos A\sin B)^2 = \cos^2 A + \cos^2 B$

$[\because \sin(A + B) = \sin A\cos B + \cos A\sin B]$

$\Rightarrow \sin^2 A\cos^2 B + \cos^2 A\sin^2 B +$

$2\sin A\sin B\cos A\cos B = \cos^2 A + \cos^2 B$

$\Rightarrow (1 - \cos^2 A)\cos^2 B + \cos^2 A(1 - \cos^2 B)$

$+2\sin A\sin B\cos A\cos B = \cos^2 A + \cos^2 B$

$\Rightarrow \cos^2 B - \cos^2 A\cos^2 B + \cos^2 A - \cos^2 A$

$\cos^2 B + 2\sin A\sin B\cos A\cos B = \cos^2 A + \cos^2 B$

$\Rightarrow 2\sin A\sin B\cos A\cos B - 2\cos 2A\cos^2 B = 0$

$\Rightarrow 2\cos A\cos B(\sin A\sin B - \cos A \cdot \cos B) = 0$

$\Rightarrow -2\cos A\cos B\cos(A + B) = 0$

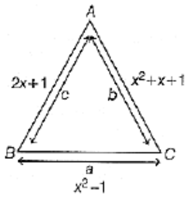
$\Rightarrow \cos A\cos B\cos(A + B) = 0$

This will be true, if

$A = \frac{\pi}{2}$ or $B = \frac{\pi}{2}$ or $A + B = \frac{\pi}{2} \Rightarrow C = \frac{\pi}{2}$

So, the triangle must be right angled triangle.

4. (d) Sides of a triangle must be positive, hence $2x + 1 > 0$ and $x^2 - 1 > 0$ and we know that $x^2 + x + 1 > 0$ for every $x \in R$. Solving these equations, we get $x > 1$.
The greatest angle of the triangle opposite to its greatest side which is $x^2 + x + 1$.



$\cos B = \frac{c^2+a^2-b^2}{2c\square}$

$\cos B = \frac{(2x+1)^2+(x^2-1)^2-(x^2+x+1)^2}{2(x^2-1)(2x+1)}$

$= \frac{-2x^3-x^2+2x+1}{2(2x^3+x^2-2x-1)} = \frac{-(2x^3+x^2-2x-1)}{2(2x^3+x^2-2x-1)} = -\frac{1}{2}$

$\cos B = -\frac{1}{2} \Rightarrow B = \frac{2\pi}{3} = 120^\circ$

5. **(a)** Area of $\triangle ABC = 10\sqrt{3}$

We know for $\triangle ABC$

Area $= \frac{1}{2}ac\sin B = \frac{1}{2}bc\sin A = \frac{1}{2}ca\sin B$

Here, $c = 8, b = 5 \Rightarrow 10\sqrt{3} = \frac{1}{2} \times 8 \times 5 \times \sin A$

$\frac{4\sqrt{3}}{8} = \sin A \Rightarrow \sin A = \frac{\sqrt{3}}{2}, A = \frac{\pi}{3}, \frac{2\pi}{3} \Rightarrow A = 60^\circ, 120^\circ$

6. **(c)** $\because (a+b+c)(a+b-c) = (a+b)^2 - c^2 = ab$

$\Rightarrow a^2+b^2-c^2+ab = 0 \Rightarrow a^2+b^2-c^2 = -ab$

$\Rightarrow \frac{2(a^2+b^2-c^2)}{2ab} = -1$

$\frac{a^2+b^2-c^2}{2ab} = -\frac{1}{2} \Rightarrow \cos C = -\frac{1}{2} \Rightarrow C = \frac{2\pi}{3}$

7. **(b)** Let $t = \frac{1}{3}s^2(s-a)(s-b)(s-c)$

$\sqrt{t} = \sqrt{\frac{s}{3}}\sqrt{s(s-a)(s-b)(s-c)}$

$\Rightarrow \sqrt{t} = \sqrt{\frac{s}{3}}\Delta \Rightarrow \sqrt{t} = \sqrt{2}\Delta$ [$\because s = 6$]

$t = 2\Delta^2$

8. **(d)** $\cos C = \frac{a^2+b^2-c^2}{2ab} = \frac{2^2+4^2-c^2}{2\times 2\times 4}$

$\cos 60^\circ = \frac{20-c^2}{16} \Rightarrow \frac{1}{2} = \frac{20-c^2}{16}$

$\frac{16}{2} = 20 - c^2 \Rightarrow c^2 = 12 \Rightarrow c = \pm 2\sqrt{3} \Rightarrow c = 2\sqrt{3}$

Now, using sin rule

$\frac{\sin 60^\circ}{2\sqrt{3}} = \frac{\sin A}{2}$

$\Rightarrow \frac{\sqrt{3}}{4\sqrt{3}} = \frac{\sin A}{2} \Rightarrow \sin A = \frac{1}{2} \Rightarrow A = 30^\circ$

In any triangle, $A + B + C = 180^\circ \Rightarrow 30^\circ + B + 60^\circ = 180^\circ$

$B = 90^\circ, A = 30^\circ, B = 90^\circ$

9. **(a)** Let $A = 2k, B = 3k, C = 7k$

In any triangle, $A + B + C = 180^\circ = \pi$

$12k = 180^\circ \Rightarrow k = 15^\circ$

A$= 30^\circ, B = 45^\circ, C = 105^\circ$

Using sine rule, $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

$\sin A : \sin B : \sin C = a : b : c$

$a : b : c = \sin 30^\circ : \sin 45^\circ : \sin 105^\circ$

$\left[\therefore \sin 105^\circ = \cos 15^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}}\right]$

$= \frac{1}{2} : \frac{1}{\sqrt{2}} : \frac{\sqrt{3}+1}{2\sqrt{2}} = \sqrt{2} : 2 : \sqrt{3}+1$

10. **(d)** cos A $= \frac{b^2+c^2-a^2}{2bc} \Rightarrow \cos 60^\circ = \frac{9+c^2-16}{6c}$

$\frac{1}{2} = \frac{c^2-7}{6c}$

$\Rightarrow c^2 - 3c - 7 = 0$

11. **(c)** $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$

$2R\sin C = c \Rightarrow \angle C = \frac{\pi}{2}$

$c = 2R\sin\frac{\pi}{2} \Rightarrow c = 2R \Rightarrow R = \frac{c}{2} \Rightarrow r = (s-c)\tan\frac{C}{2}$

$\Rightarrow r = (s-c) \Rightarrow 2R + 2r$

$= \frac{2c}{2} + 2(s-c) = 2s - 2c + c$

$= a + b + c - c = a + b$

$\Delta = a^2 - (b-c)^2$

$\Delta = a^2 - b^2 - c^2 + 2bc = 2bc - (b^2+c^2-a^2)$

$= 2bc - 2bc\frac{(b^2+c^2-a^2)}{2bc}$

12. (b) $\Delta = a^2 - (b-c)^2$

$$\Delta = a^2 - b^2 - c^2 + 2bc = 2bc - (b^2 + c^2 - a^2)$$

$$= 2bc - 2bc\frac{(b^2+c^2-a^2)}{2bc}$$

$$\cos A = \frac{(b^2+c^2-a^2)}{2bc}$$

$$\Delta = 2bc(1-\cos A) = 4bc\sin^2\frac{A}{2}$$

Also, $\Delta = \frac{bc}{2}\sin A = bc\sin\frac{A}{2}\cos\frac{A}{2}$

On dividing Eq. (i) by Eq. (ii), we get

$$1 = \frac{4bc\sin^2\frac{A}{2}}{bc\sin\frac{A}{2}\cos\frac{A}{2}} \Rightarrow \tan\frac{A}{2} = \frac{1}{4}$$

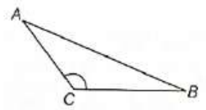
$$\tan A = \frac{2\tan\frac{A}{2}}{1-\tan^2\frac{A}{2}} = \frac{2\left(\frac{1}{4}\right)}{1-\frac{1}{16}} = \frac{\frac{1}{15}}{\frac{1}{16}} = \frac{8}{15} \Rightarrow \tan A = \frac{8}{15}$$

13. (d) Let a and b be the sides of triangle

$$a + b = x \Rightarrow ab = y$$

Given, $x^2 - z^2 = y$

Here, z is its third side.



$$(a+b)^2 - z^2 = ab \Rightarrow a^2 + b^2 + 2ab - z^2 = ab$$

$$a^2 + b^2 + ab = z^2 \Rightarrow a^2 + b^2 - z^2 = -ab$$

$$\frac{a^2+b^2-z^2}{2ab} = -\frac{1}{2} \Rightarrow \cos C = \frac{2\pi}{3}$$

So, the triangle is obtuse angled.

14. (a) $a\left(\frac{1+\cos C}{2}\right) + c\left(\frac{1+\cos A}{2}\right) = \frac{3b}{2}$

$$\left[\because \cos^2\frac{\theta}{2} = \frac{1+\cos\theta}{2}\right]$$

$$\Rightarrow a + a\cos C + c + c\cos A = 3b$$

$$\Rightarrow a + c + (a\cos C + c\cos A) = 3b$$

Using projection formula,

$$a + c + b = 3b \Rightarrow 2b = a + c$$

Hence, the sides of triangle are in AP.

15 (a) $\tan\left(\frac{B-C}{2}\right) = \frac{1}{3}\tan\left(\frac{B+C}{2}\right)$

Using Napier's analogy, $\tan\left(\frac{B-C}{2}\right) = \left(\frac{b-c}{b+c}\right)\cot\frac{A}{2}$

As we know, $\tan\left(\frac{B+C}{2}\right) = \cot\left(\frac{A}{2}\right)$

$$= \frac{1}{3}\tan\left(\frac{B+C}{2}\right) = \frac{1}{3}\cot\frac{A}{2} = \frac{b-c}{b+c}\cot\frac{A}{2}$$

Hence, $\frac{b-c}{b+c} = \frac{1}{3} \Rightarrow b = 2c \Rightarrow b:c = 2:1$

16. (a) Let the side of the triangle are a, b, c. It is given that the perimeter of a $\triangle ABC$ is 6 times the arithmetic mean of the sines of its angles.

$$\therefore\ a + b + c = 6\left(\frac{\sin A + \sin B + \sin C}{3}\right)$$

$$a + b + c = 2(\sin A + \sin B + \sin C) \quad \text{..... (i)}$$

From the law of sine, $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$

$$\Rightarrow\ a = k\sin A \Rightarrow b = k\sin B \text{ and } c = k\sin C$$

$$\therefore a + b + c = k(\sin A + \sin B + \sin C) \quad \text{... (ii)}$$

Hence, $k = 2$

$$\Rightarrow a = 2\sin A \Rightarrow 1 = 2\sin A \Rightarrow \sin A = \frac{1}{2} \Rightarrow A = \frac{\pi}{6}$$